

Jay Patel

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Summary

Full-Stack Developer (MERN & Python/FastAPI) with production experience building AI-powered applications using LLMs, RAG, LangChain, and LangGraph. Skilled in REST API design, relational and NoSQL databases, async job queues, and CI/CD. Experienced across the full lifecycle: authentication and authorization, database indexing and query optimization, and deployment of full-stack systems to production.

Skills

Languages: JavaScript, TypeScript, Python, C++, SQL

Backend & APIs: Node.js, Express.js, FastAPI, SQLAlchemy, REST APIs, WebSockets, Microservices

AI / ML: LLMs, RAG (Retrieval-Augmented Generation), LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Databases, MCP, Voice APIs, Basic ML

Databases: PostgreSQL, MongoDB, MySQL, Redis, pgvector

System Design: System Design, Low-Level Design (LLD), Kafka, RabbitMQ, BullMQ

Testing: Jest, Supertest, Pytest, Unit Testing, API Testing

DevOps: Docker, GitHub Actions, AWS, CI/CD

Frontend: React.js, Next.js, Tailwind CSS

Core CS: Data Structures & Algorithms, DBMS, Operating Systems, OOP

Experience

AI / MERN Stack Intern

Sep 2026 – Present

Techsteck Solutions

- Developing and maintaining backend services for a B2B Taxi Marketplace Platform using PHP, Laravel, REST APIs, and PostgreSQL, supporting core booking and marketplace workflows.
- Collaborating on API design and database schema decisions to support multi-tenant marketplace operations across driver, rider, and admin roles.

Full Stack Developer Intern

Jan 2026 – Jun 2026

Agevole Innovation Pvt. Ltd.

- Built and deployed a full-stack e-commerce platform on the MERN stack, implementing the complete authentication and authorization flow (JWT, RBAC) across frontend, backend, and database layers.
- Designed and implemented REST APIs and CRUD operations for product, cart, and order management, taking the platform from development through to production deployment.

Projects

Newsbit – Python, FastAPI, PostgreSQL, SQLAlchemy, Next.js, LLMs, LangGraph

newsbit.in | [GitHub](#)

- Architected a LangGraph tool-calling news agent (FastAPI + WebSockets) with 4-way LLM failover across models/API keys, routing between internal metadata/vector search and a live web-search fallback to stay current without over-calling the LLM.
- Built a hybrid retrieval pipeline combining entity-based metadata search (alias/entity canonicalization, e.g. Meta ↔ Facebook, Alphabet ↔ Google) with pgvector cosine-similarity semantic search, ranked by entity match count and recency.
- Designed an async, timezone-aware cron pipeline (APScheduler) orchestrating ingestion, popularity ranking, LLM summarization, TTS audio pre-warming, and embedding backfill as isolated stages, so a single stage's failure (e.g. rate limits) never blocks the rest.

PromptX – Node.js, Express.js, PostgreSQL, MongoDB, BullMQ, Docker

promptx.co.in | [GitHub](#)

- Built a production-ready AI prompt platform with a modular Express backend, dual MongoDB/PostgreSQL persistence, and BullMQ/Redis-backed async job queues with a dedicated worker process for AI prompt enhancement and multi-file context processing.
- Built a retrieval-augmented (RAG) AI assistant with a custom relevance-ranking retrieval pipeline that grounds Groq/OpenAI LLM responses in site-specific context, reducing hallucinated answers.
- Implemented JWT authentication with short-lived access tokens, TTL-indexed refresh tokens, and Google OAuth; eliminated a full collection scan on the conversation-history API via a compound MongoDB index, cutting documents examined per query by ~99.9% (50,023 → 50) at 50k-message scale.

Education

G.H. Patel College of Engineering and Technology (GCET), Anand

Bachelor of Technology (B.Tech.) in Information Technology

2022 – 2026

GPA: 7.32 / 10